MARK SUSOL

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PROFESSIONAL SUMMARY

Principal AI Architect and engineering leader specializing in designing and deploying multi-agent AI systems at enterprise scale. Currently directing the AI product architecture for Mitratech's HR Solutions division, defining the vision and execution strategy for secure, extensible, and GenAI-powered platforms supporting global legal and HR workflows. Formerly at Oracle, led the architecture and implementation of OCI Generative AI Agents and Model Context Protocol (MCP) server infrastructure powering Oracle's internal AskOracle search and support experiences. Prior experience at Google building advanced NLP and MLOps platforms that improved model routing accuracy by 35% and scaled analytics, engagement scoring, and clickstream insights across products.

WORK HISTORY

MITRATECH, Austin, TX (remote) - **Principal AI Architect** **Feb 2026 - Current**

- Direct the AI product architecture for Mitratech's HR Solutions division, aligning multi-agent orchestration, retrieval-augmented generation (RAG), and data privacy frameworks to deliver scalable, compliant, and explainable AI capabilities across global HR platforms.

ORACLE, Broomfield, CO (remote) - **Principal Software Engineer** **Jun 2024 - Feb 2026**

- Architected OCI Generative AI agent-based workflows for Oracle's AskOracle search experience, integrating RAG, multi-tool capabilities, and contextual intent routing across internal knowledge bases.
- Designed and implemented MCP (Model Context Protocol) servers that orchestrate specialized agents for search, analytics, and operational troubleshooting, exposing structured APIs and data contracts for microservice consumption.
- Led the evolution of data ingestion pipelines from monolithic batch jobs to Kubernetes-native, DAG-driven architectures (via Argo Workflows), improving throughput, resilience, and cloud cost efficiency.
- Developed a denormalized, role-aware user permission model for vector indexing and RAG retrieval of restricted enterprise documents, enabling secure, fine-grained access control within AI systems.
- Partnered with data engineering, platform, and security teams to define agent schemas and streaming-optimized data contracts for low-latency, event-driven AI workflows on OCI.
- Mentored engineers in GenAI architecture, observability, and deployment best practices for scalable agentic systems.

UNIVERSITY OF COLORADO BOULDER - Leeds School of Business, - Adjunct Lecturer **Aug 2024 - Dec 2024**

- BAIM 4200 – Advanced Business Analytics, a ML based lab course using Google Colab.

GOOGLE, Boulder, CO - **Staff Data Scientist** **Nov 2018 - Jan 2024**

- Collaborated with product and research teams to analyze real-world business problems across various domains including Google's supply chain, user support, NLP research, and generative AI research.
- Applied predictive ML specifically using NLP to develop new business strategies and solutions to address complex challenges like detection of emerging issues, and server deployment failures.
- Improved Google's support routing match accuracy by 35% with predictive online ML models using NLP-based

TensorFlow models, saving Google substantial costs via real-time serving technologies.
- Provided qualitative analysis for Google's Text to Image Generative AI initiative, providing valuable insights regarding benchmarking internal tools performance.
- Developed end-to-end AI/ML workflows primarily with Python on Google's Tensorflow framework tailored to address our specific business challenges.
- Spearheaded the creation of user engagement scoring models, focusing on metrics such as effort scores, to gauge customer engagement levels across Google's diverse support ecosystems and individual products.
- Collaborate with cross-functional teams to ensure the development of comprehensive scoring models that accurately reflect user behaviors and interactions across platforms.
- Managed analytical projects with cross-functional stakeholders, documenting project requirements and design principles, and communicating routine progress and milestones met.
- Maintained custom Python and SQL modules deployed in NLP AI/ML workflows across multiple domains through rigorous unit and integration testing.

GOOGLE, Mountain View, CA - **SQL Readability Admin** **Mar 2018 - Nov 2023**

- Led a cross-functional admin team at Google, analyzing GoogleSQL features and streamlining the SQL style guide for improved readability across the organization.
- Improved SQL code optimization by mentoring Googlers in code reviews, resulting in consistently ranking Top 5 for review completions.
- Demonstrated strong instructional skills and subject expertise by achieving a high feedback score of 4.7/5 for training 60 Googlers per quarter in GoogleSQL Functions and Modules.

GOOGLE, Mountain View, CA - **Senior Data Engineer** **Sep 2015 - Oct 2018**

- Orchestrated the design and implementation of robust data integration strategies and architectural frameworks, focusing on raw log processing and the integration of support interaction events.
- Established and led the gUP Analytics Tech Working Group, enhancing the team's methodologies and overall workflow through dedicated expertise.
- Conducted 193 technical interviews evaluating candidates for data engineering or analyst roles.

COGNIZANT, Mountain View, CA - **Technical Solutions Engineer** **Sep 2013 - Aug 2015**

- Created Business Intelligence tools using Google App Engine, integrated with Google Data Services.
- Improved data processing time by incorporating MapReduce in Python and Go, in ETL processes.

REI SYSTEMS, Reston, VA - **Software Engineer** **Feb 2012 - Aug 2013**

- Developed SaaS predictive analytics software in Python; Logistic/Linear Regression.
- Built interactive visualizations using AJAX, Drupal REST API, HighCharts, and D3.js

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EDUCATION

Master of Science, Applied Physics – University of Maryland Baltimore County • Baltimore, MD (**1995**)

Bachelor of Arts, Physics – University of Maryland Baltimore County • Baltimore, MD (**1992**)

AWARDS

VP Award: Scaled tracking of gPay product outcomes following support interactions. | **Google (2018)**

VP Award: Design, development, and launch of clickstream data infrastructures. | **Google** (2017)